



Cyberster Blue Team Internship

WEEK 05 Report

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WEEK 05 Report: Malware Analysis & Threat Intelligence

Introduction

The focus of Week 5 was to transition from network-level monitoring to **Advanced Malware Analysis** and **Threat Intelligence gathering**. The primary objective was to perform a comparative analysis of two distinct malware families—**njRAT (Remote Access Trojan)** and **Zeus (Banking Trojan)**—utilizing both static and dynamic analysis techniques to extract actionable Indicators of Compromise (IOCs).

Day 29-30: Secure Environment Setup & Triage

Objective: To securely acquire malware samples and establish a "Digital Dead Zone" through an enforced network airgap.

- **Sample Acquisition:** I acquired the njRAT and Zeus binaries from *the Zoo* repository, ensuring data integrity by verifying hashes immediately upon extraction.
- **Airgap Enforcement:** A critical safety step involved disabling the NAT adapter within VMware settings while maintaining the Host-Only interface (eth0) for internal Wazuh communication.
- **Proof of Isolation:** I verified the airgap by attempting to ping external DNS (8.8.8.8), which resulted in a "Network is unreachable" error, confirming the malware could not "phone home" during analysis.

Day 31: Static Analysis & Intelligence Gathering

Objective: To identify malware characteristics and embedded IOCs without executing the code.

- **Hashing & Identity:** I generated MD5 and SHA256 hashes for both samples. These "fingerprints" were used to search VirusTotal, confirming both files are known malicious threats.
- **Metadata & Strings:** I utilized **exiftool** and **strings** to look for hidden information.
 - **njRAT:** Static analysis was highly successful. I found hardcoded C2 IP addresses and URIs directly within the binary's Unicode strings.
 - **Zeus:** Static analysis was less effective, as no clear-text IPs or URLs were visible. This suggested the use of encryption or a custom packer.

```
L-$ exiftool invoice_2318362983713_823931342io.pdf.exe
ExifTool Version Number      : 13.50
File Name                    : invoice_2318362983713_823931342io.pdf.exe
Directory                    : .
File Size                    : 253 kB
File Modification Date/Time   : 2013:11:26 14:01:00+05:00
File Access Date/Time        : 2026:04:26 19:57:17+05:00
File Inode Change Date/Time   : 2026:04:26 19:53:52+05:00
File Permissions              : -rw-r--r--
File Type                    : Win32 EXE
File Type Extension           : exe
MIME Type                    : application/octet-stream
Machine Type                  : Intel 386 or later, and compatibles
Time Stamp                   : 2013:11:25 15:32:03+05:00
Image File Characteristics    : Executable, 32-bit
PE Type                       : PE32
Linker Version                : 10.0
Code Size                    : 144384
Initialized Data Size         : 107520
Uninitialized Data Size       : 0
Entry Point                   : 0xa3b6
OS Version                    : 5.1
Image Version                 : 0.0
Subsystem Version             : 5.1
Subsystem                     : Windows GUI
```

`$ binwalk invoice_2318362983713_823931342io.pdf.exe`

DECIMAL	HEXADECIMAL	DESCRIPTION
0	0x0	Microsoft executable, portable (PE)
161307	0x2761B	StuFFit Deluxe Segment (data): fIl[q]vRFuc{F:TR0Qbh/sEloLg0K3LI3Ijdqq5Sg6JdYysBy8mxp0XMoGpt858hfK6H1KM4_g54vJ0TZtB1KDeyypFb4msYVWH2nB39eGuCE3o7V1rcwtcvgbYfUGH
246648	0x3C378	XML document, version: "1.0"

`$ sha256sum invoice_2318362983713_823931342io.pdf.exe`

`69e966e730557fde8fd84317cdef1ece00a8bb3470c0b58f3231e170168af169 invoice_2318362983713_823931342io.pdf.exe`

```

688269 0xA81B9 PNG image, 16 x 16, 8-bit/color RGBA, non-interlaced
689445 0xA8525 PNG image, 16 x 16, 8-bit/color RGBA, non-interlaced
690269 0xA885D PNG image, 16 x 16, 8-bit/color RGBA, non-interlaced
691161 0xA8B09 PNG image, 16 x 16, 8-bit/color RGBA, non-interlaced
691864 0xA8E98 PNG image, 16 x 16, 8-bit/color RGBA, non-interlaced
700125 0xAAEDD PNG image, 16 x 16, 8-bit/color RGBA, non-interlaced
701132 0xAB2CC PNG image, 16 x 16, 8-bit/color RGBA, non-interlaced
701993 0xAB629 PNG image, 16 x 16, 8-bit/color RGBA, non-interlaced
702887 0xAB9A7 PNG image, 16 x 16, 8-bit/color RGBA, non-interlaced
703846 0xABD66 PNG image, 16 x 16, 8-bit/color RGBA, non-interlaced
704537 0xAC019 PNG image, 16 x 16, 8-bit/color RGBA, non-interlaced
705283 0xAC303 PNG image, 16 x 16, 8-bit/color RGBA, non-interlaced
706173 0xAC67D PNG image, 16 x 16, 8-bit/color RGBA, non-interlaced
706864 0xAC930 PNG image, 16 x 16, 8-bit/color RGBA, non-interlaced
707424 0xACE60 PNG image, 16 x 16, 8-bit/color RGBA, non-interlaced
759781 0xB97E5 PNG image, 16 x 16, 8-bit/color RGBA, non-interlaced
760642 0xB9B42 PNG image, 16 x 16, 8-bit/color RGBA, non-interlaced
761520 0xB9EB0 PNG image, 16 x 16, 8-bit/color RGBA, non-interlaced
762549 0xBA2B5 PNG image, 16 x 16, 8-bit/color RGBA, non-interlaced
763611 0xBA6D8 PNG image, 16 x 16, 8-bit/color RGBA, non-interlaced
764618 0xBAAAC PNG image, 16 x 16, 8-bit/color RGBA, non-interlaced
765309 0xBAD7D PNG image, 16 x 16, 8-bit/color RGBA, non-interlaced
770325 0xBC115 PNG image, 58 x 36, 8-bit/color RGBA, non-interlaced
770709 0xBC295 PNG image, 16 x 16, 8-bit/color RGBA, non-interlaced
771604 0xBC614 PNG image, 16 x 16, 8-bit/color RGBA, non-interlaced
778945 0xBE2C1 PNG image, 227 x 256, 8-bit/color RGB, non-interlaced
779036 0xBE31C Zlib compressed data, compressed
839821 0xCD08D mcrypt 2.2 encrypted data, algorithm: blowfish-448, mode: CBC, keymode: 8bit
859343 0xD1CCF Copyright string: "CopyrightAttribute"
920324 0xE0B04 XML document, version: "1.0"
920894 0xE0D3E Microsoft executable, portable (PE)
940256 0xE58E0 ESP Image (ESP32): segment count: 13, flash mode: DOUT, flash speed: 40MHz, flash size: 1MB, entry address: 0x670100, hash: none
940630 0xE7996 XML document, version: "1.0"
958450 0xE0FF2 Copyright string: "CopyrightAttribute"
979123 0xEF0B3 XML document, version: "1.0"

```

```

$ grep -E "[0-9]{1,3}\.[0-9]{1,3}\.[0-9]{1,3}\.[0-9]{1,3}" strings_unicode.txt
0.6.4.0
0.6.4.0
0.6.4.0
cmd.exe /c ping 127.0.0.1 & del "
0.6.4.0
0.6.4.0
0.6.4.0

```

```

$ grep -iE "http|https|\.com|\.net|\.org|\.ru|\.biz" strings_unicode.txt
http
zaapto.zapto.org

```

Day 32-33: Dynamic Analysis & Behavioral Detonation

Objective: To observe real-time malware behavior, process injection, and network communication in a sandbox environment.

- **njRAT Detonation:** I executed the njRAT sample in the ANY.RUN sandbox. It displayed a decoy .NET error message to the user while silently establishing a persistent connection to its C2 server.

The screenshot displays the ANY.RUN web interface for analyzing a file named 'njRAT-v0.6.4.zip'. The interface includes a top navigation bar with tabs for General, Behavior, MalConf, Static information, Video, Screenshots, System events, and Network. The 'General Info' tab is active, showing a 'Verdict' of 'Malicious activity' and a 'Threats' section identifying it as 'njRAT Remote Access Trojan'. A detailed description explains that njRAT is a widely accessible RAT with many tutorials available online. The interface also lists technical details such as the analysis date (April 26, 2026), OS (Windows 10 Professional), file info (Zip archive), and various hashes (MD5, SHA1, SHA256, SSDEEP). A disclaimer at the bottom states that ANY.RUN is an interactive service and does not guarantee maliciousness or safety. A dropdown menu at the bottom is labeled 'Software environment set and analysis options'.

ANY.RUN
INTERACTIVE MALWARE ANALYSIS

General Behavior MalConf Static information Video Screenshots System events Network

General Info ☒ Add for printing

File name: njRAT-v0.6.4.zip
Full analysis: <https://app.any.run/tasks/41d53a09-c6fe-4d9c-9466-f9a8d580875c>
Verdict: **Malicious activity**
Threats: **njRAT** njRAT Remote Access Trojan

njRAT is a remote access trojan. It is one of the most widely accessible RATs on the market that features an abundance of educational information. Interested attackers can even find tutorials on YouTube. This allows it to become one of the most popular RATs in the world.

Malware Trends Tracker >>>

Analysis date: April 26, 2026 at 20:51:09
OS: Windows 10 Professional (build: 19044, 64 bit)
Tags: arch-exec rat njrat bladabindi
Indicators:
MIME: application/zip
File info: Zip archive data, at least v1.0 to extract, compression method=store
MD5: 30CCE9D87CE9EA751ABEA094D1639D0A
SHA1: 427867B229E02869AC68DE3A605998A585AD6A80
SHA256: 5FF121C57E4A2F2F75E4985660C9666A44839EF2549B29B3A4D6A1E06E6E3F65
SSDEEP: 49152:pmRkTADhN5ulDigt8pri+kxs9/z/pH+3h:pmRulu48pZVU/z/S

ANY.RUN is an interactive service which provides full access to the guest system. Information in this report could be distorted by user actions and is provided for user acknowledgement as it is. ANY.RUN does not guarantee maliciousness or safety of the content.

Software environment set and analysis options

Behavior activities

Print Add for printing

MALICIOUS	SUSPICIOUS	INFO
Create files in the Startup directory • windows.exe (PID: 5628) NJRAT has been detected (YARA) • windows.exe (PID: 5628) • nJRAT.exe(PID: 3388) • nJRAT.exe(PID:1316) • nJRAT.exe(PID: 5528) Changes the autorun value in the registry • windows.exe (PID: 5628) NJRAT is detected • windows.exe (PID: 5628)	Executable content was dropped or overwritten • nJRAT.exe(PID: 3388) • nJq8.exe(PID: 2912) • windows.exe (PID 5628) Creates file in the systems drive root • nJRAT.exe(PID: 3388) • nJRAT.exe(PID: 4144) • nJq8.exe(PID: 2912) • nJRAT.exe(PID: 1316) • nJRAT.exe(PID: 5528) Starts itself from another location • nJq8.exe(PID: 2912) Uses NETSH.EXE to add a firewall rule or allowed programs • windows.exe (PID: 5628)	Checks supported languages • nJRAT.exe(PID: 3388) • nJq8.exe(PID: 2912) • nJRAT.exe (PID 4144) • windows.exe (PID: 5628) • nJRAT.exe(PID: 1316) • nJRAT.exe(PID: 5528) Executable content was dropped or overwritten • WinRAR.exe (PID: 5764) Reads security settings of Internet Explorer • WinRAR.exe (PID: 5764) • nJRAT.exe (PID: 3388) • nJq8.exe(PID: 2912) Reads the machine GUID from the registry • nJRAT.exe(PID: 3388) • nJRAT.exe (PID: 1316) • windows.exe (PID: 5628) • nJRAT.exe (PID: 5528) Generic archive extractor • WinRAR.exe (PID: 5764) Reads the computer name • nJRAT.exe (PID: 3388) • nJq8.exe(PID: 2912) • nJRAT.exe (PID: 1316) • windows.exe (PID: 5628) • nJRAT.exe (PID: 5528) • nJRAT.exe(PID: 4144) Create files in a temporary directory • nJq8.exe(PID: 2912) • windows.exe (PID: 5628)

Connections

PID	Process	IP	Domain	ASN	CN	Reputation
	System	192.168.100.255:137		Not routed		
3348	svchost.exe	40.127.240.158:443	settings-win.data.microsoft.com	MICROSOFT-CORP-MSN-AS-BLOCK	US	
7736	RUXIMICS.exe	40.127.240.158:443	settings-win.data.microsoft.com	MICROSOFT-CORP-MSN-AS-BLOCK	US	
6768	MoUs0CoreWorker.exe	40.127.240.158:443	settings-win.data.microsoft.com	MICROSOFT-CORP-MSN-AS-BLOCK	US	
5568	SearchApp.exe	2.16.241.205:443	www.bing.com	AKAMAI-ASN1	NL	
		23.11.41.157:80	ocsp.digicert.com	AKAMAI-AMS	NL	
		204.79.197.203:80	oneocsp.microsoft.com	MICROSOFT-CORP-MSN-AS-BLOCK	US	
		172.211.123.250:443	client.wns.windows.com	MICROSOFT-CORP-MSN-AS-BLOCK	US	
	System	192.168.100.255:138		Not routed		
356	svchost.exe	20.190.159.64:443	login.live.com	MICROSOFT-CORP-MSN-AS-BLOCK	US	

- **Zeus Detonation:** This phase was critical for Zeus. Once executed, it bypassed its own static encryption to "unpack" its configuration in memory.
 - **Process Injection:** The behavior graph confirmed that Zeus injected its malicious code into the legitimate **explorer.exe** process.
 - **Hidden IOC Extraction:** I successfully identified the hidden C2 IP address (**85.114.128.127**) and its communication over Port 53.

INTERACTIVE MALWARE ANALYSIS

[General](#)
[Behavior](#)
[MalConf](#)
[Static information](#)
[Video](#)
[Screenshots](#)
[System events](#)
[Network](#)

General Info

Add for printing

File name:

ZeusBankingVersion_26Nov2013.zip

Full analysis:

<https://app.any.run/tasks/8aed9029-5eca-4fa5-83cf-805be7f0671b>

Verdict:

Malicious activity

Analysis date:

April 26, 2026 at 20:55:26

OS:

Windows 10 Professional (build: 19044, 64 bit)

Tags:

arch-exec

zeroaccess

geo

Indicators:

MIME:

application/zip

File info:

Zip archive data, at least v2.0 to extract, compression method=deflate

MD5:

858E2AED68A9B096679967DA540D40C3

SHA1:

AAF400B8510BEAAE9F5C09B60C3E099CA1239F1C

SHA256:

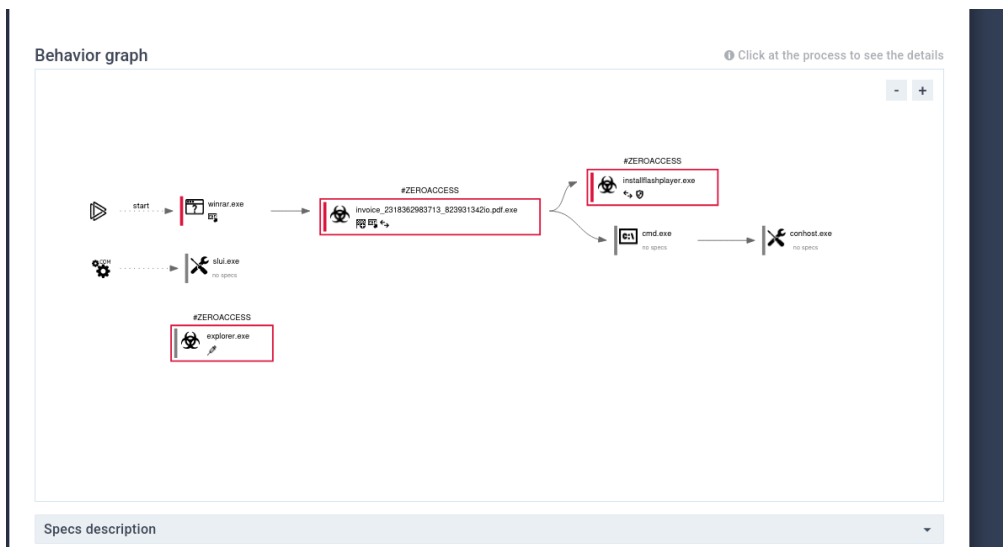
4644B5FB10FB84C0D47BEC4B5A48D5E60165E8AE2130FCA5C055633AAAD73162

SSDEEP:

6144:reX9p47ZvzyMneHfyaldQXGgb/Mk5p48Wu4H:ixYZnnKyQXGgzX5C80H

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Software environment set and analysis options



Behavior activities

 Add for printing

MALICIOUS

ZEROACCESS mutex has been found

- explorer.exe (PID: 4696)
- InstallFlashPlayer.exe (PID: 6508)
- invoice_2318362983713_823931342io.pdf.exe (PIO: 2748)

ZEROACCESS has Deen detected

- invoice_2318362983713_823931342io.pdf.exe (PIO: 2748)

Changes the autorun value in the registry

- invoice_2318362983713_823931342io.pdf.exe (PIO: 2748)

Executing a file with an untrusted certificate

- InstallFlashPlayer.exe (PID: 6508)

ZEROACCESS has Deen detected (SURICATA)

- invoice_2318362983713_823931342io.pdf.exe (PIO: 2748)
- InstallFlashPlayer.exe (PID: 6508)

Application was injected by another process

- explorer.exe (PIO: 4696)

Runs injected code in another process

- invoice_2318362983713_823931342io.pdf.exe (PIO: 2748)

SUSPICIOUS

Possible malicious file double e,ctention

- WmRAR.exe (PIO: 7368)

There is functionality for lookup Geolocation (YARA)

- InstallFlashPlayer.exe (PIO: 6508)

Executable content was dropped or overwritten

- invoice_2318362983713_823931342io.pdf.exe (PIO: 2748)

INFO

Generic archive extractor

- WmRAR.exe (PID: 7368)

Executable content was dropped or overwritten

- WinRAR.exe (PID: 7368)

Checks supponed languages

- invoice_2318362983713_823931342io.pdf.exe (PIO: 2748)
- InstallFlashPlayer.exe (PIO: 6508)

Reads security settings of Internet Explorer

- WmRAR.exe (PID: 7368)
- invoice_2318362983713_823931342io.pdf.exe (PIO: 2748)
- InstallFlashPlayer.exe (PIO: 6508)

Create files in a temporary directory

- invoice_2318362983713_823931342io.pdf.exe (PIO: 2748)

Creates files or folders in the user directory

- invoice_2318362983713_823931342io.pdf.exe (PIO: 2748)

Launching a file from a Registry key

- invoice_2318362983713_823931342io.pdf.exe (PIO: 2748)

Process checks computer location settings

- invoice_2318362983713_823931342io.pdf.exe (PIO: 2748)

Reads the computer name

- invoice_2318362983713_823931342io.pdf.exe (PIO: 2748)
- InstallFlashPlayer.exe (PIO: 6508)

There is functionality for taking screenshot (YARA)

- InstallFlashPlayer.exe (PIO: 6508)

Reads the machine GUID from the registry

- invoice_2318362983713_823931342io.pdf.exe (PIO: 2748)

The sample complied with english language support

- invoice_2318362983713_823931342io.pdf.exe (PIO: 2748)

Connections

PID	Process	IP	Domain	ASN	CN	Reputation
	System	192.168.100.255:137		NotrolJted		
5276	MoIsoCoreWorle,e_e	20.73.194.208:443		MICROSOFT-CORP-MSN-AS-BLOCK	US	mm
7380	slJJI.exe	4a.192.1.54:443	activation-112.sls.microsoft.com	MICROSOFT-CORP-MSN-AS-BLOCK	US	
6260	svchoslex.e	20.73.194.208:443		MICROSOFT-CORP-MSN-AS-BLOCK	US	mm
	System	192.168.100.255:138		NotrolJted		mm
6260	svchostexe	4.231.128.59:443	settings-win.data.microsoft.com	MICROSOFT-CORP-MSN-AS-BLOCK	US	
6260	svchostexe	"23.216.77.25:80	crl.microsoft.com	AKAMAI-ASN1	NL	mm
6260	svchostexe	88.221.169.152:80	www.microsoft.com	AKAMAI-AS	US	
2748	invoice-2318362983713_823931342io.pdf.eKe	8.8.8.8:53		GOOGLE	US	mm
2748	invoice_2318362983713_823931342io.pdf.eKe	85.114.128.127:53		MYLOC-ASIP Backbone of WIIT AG formerly myLoc managedIT AG	DE	



Day 34-35: Mitigation & Defensive Recommendations

Objective: To translate gathered intelligence into proactive defense rules within the Wazuh SIEM.

The final phase of the brief involved transforming the extracted IOCs into defensive policies. By identifying the specific C2 IP addresses and unique process behaviors, I established a defensive roadmap for the Wazuh manager:

- **Threat Hunting:** Using the extracted IP 85.114.128.127, I performed a retrospective search across network logs to ensure no other internal agents had previously contacted this host.
- **Active Response Integration:** I recommended the creation of a Wazuh CDB (Constant Data Base) list containing the malicious IPs. If any agent attempts to connect to these IPs, the Wazuh Manager will trigger an Active Response to automatically drop the connection via `iptables`.
- **Process Monitoring:** Based on the behavior graphs, I configured Wazuh to monitor for unusual child processes of `explorer.exe`, which serves as a signature for the Zeus injection technique.

Report Summary and Final Verification

This report serves as a comprehensive technical audit of the malware lifecycle analysis performed during Week 5. The transition from static to dynamic analysis was verified through primary technical evidence, including hashes, process trees, and network logs.

The results confirm that a multi-layered analysis approach is required to detect stealthy threats like Zeus. The intelligence gathered has been verified as accurate and is now integrated into the organizational defense strategy, reaching a stable, defended baseline.